

DTSA 5706

Measurement Systems Analysis

FORMAT	CATEGORY	CREDITS
Online (DTSA)	Elective	1
SECTION	BEST ROLE MATCH	AVG ALIGNMENT
Section 4 – Elective	ML Engineer (66.3 / 100)	51.5 / 100
SPECIALIZATION Statistical Quality Improvement		

52

CURRENT AVG SCORE
across 6 job roles

86

POTENTIAL SCORE
with recommended additions

+43

MAX SCORE GAIN
best single-role improvement

CURRENT COURSE DESCRIPTION

Students will learn to analyze measurement systems for process stability and statistical control and why having a stable measurement process is imperative prior to performing any statistical analysis. Students will analyze continuous measurement systems and statistically characterize both accuracy and precision using R software. Students will perform measurement systems analysis for potential, short term and long term statistical control and capability.

INDUSTRY ALIGNMENT BY JOB ROLE

JOB ROLE	CURRENT SCORE	PROJECTED SCORE	POINT GAIN	PRIORITY
ML Engineer	66.3	87.0	+20.7	MEDIUM
Data Scientist	61.0	100.0	+39.0	HIGH
Machine Learning Engineer	58.2	94.8	+36.5	HIGH
Applied Scientist	49.3	79.8	+30.5	HIGH
AI Engineer	45.4	81.2	+35.8	HIGH
MLOps Engineer	29.0	71.6	+42.6	HIGH

RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold (**example**) appear as gaps across 3 or more job roles.

Model Deployment & Serving 2 of 6 roles

production

LLMs & Generative AI 1 of 6 roles

storage

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS

- 1 **MEDIUM** **Consider Model Deployment & Serving content** — relevant to 2 industry roles. Could be introduced via case studies or optional readings. Key terms: production.
- 2 **LOW** **Optional: LLMs & Generative AI** — would broaden appeal to one additional role. Candidate terms: storage.

GAP KEYWORDS BY ROLE

ML Engineer 66.3 → 87.0 (+20.7 pts)

media data learning machine models model production engineering

Data Scientist 61.0 → 100.0 (+39.0 pts)

rules data sets characteristics presentation evolution simply different

Machine Learning Engineer 58.2 → 94.8 (+36.5 pts)

data predictive engineering sets learning tools solutions exploratory

Applied Scientist 49.3 → 79.8 (+30.5 pts)

storage problems models technical data learning engineering design

AI Engineer 45.4 → 81.2 (+35.8 pts)

data engineering models learning build technical model based

MLOps Engineer 29.0 → 71.6 (+42.6 pts)

detection data model production learning machine media image

Methodology: Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum-Industry Alignment project team.